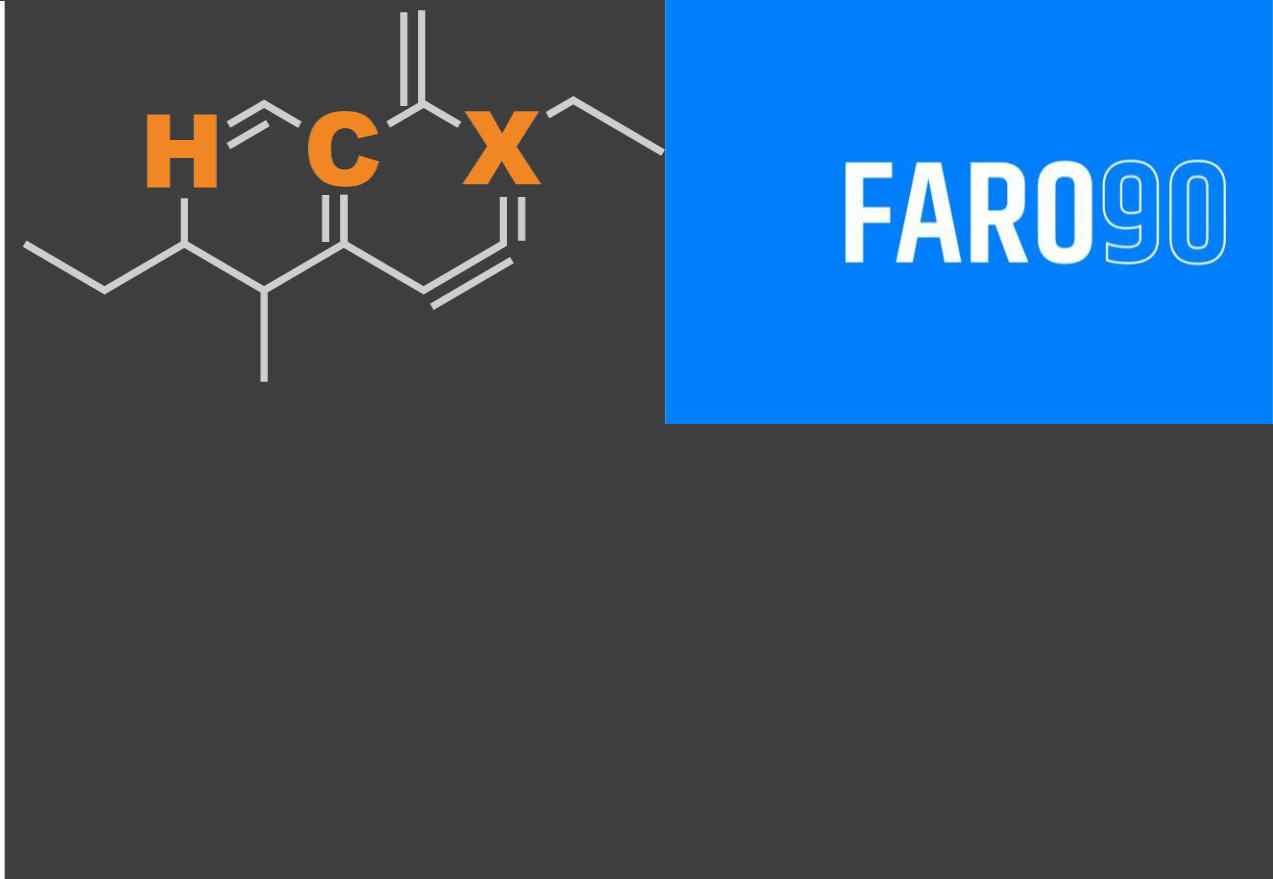




Ethanol Blending in Gasoline - Malta

September, 2025

Ethanol Blending in Europe

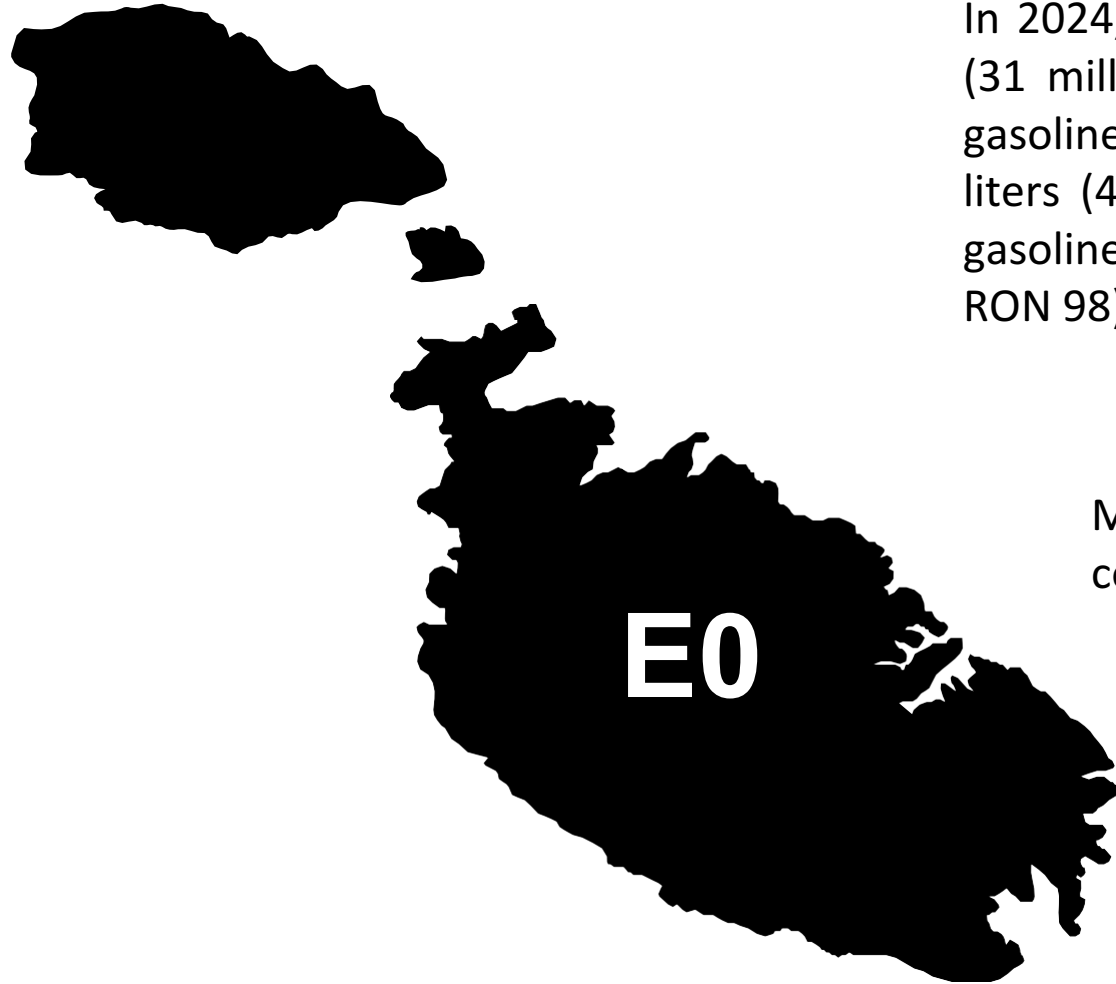
There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Ethanol Gasoline Blending in Malta

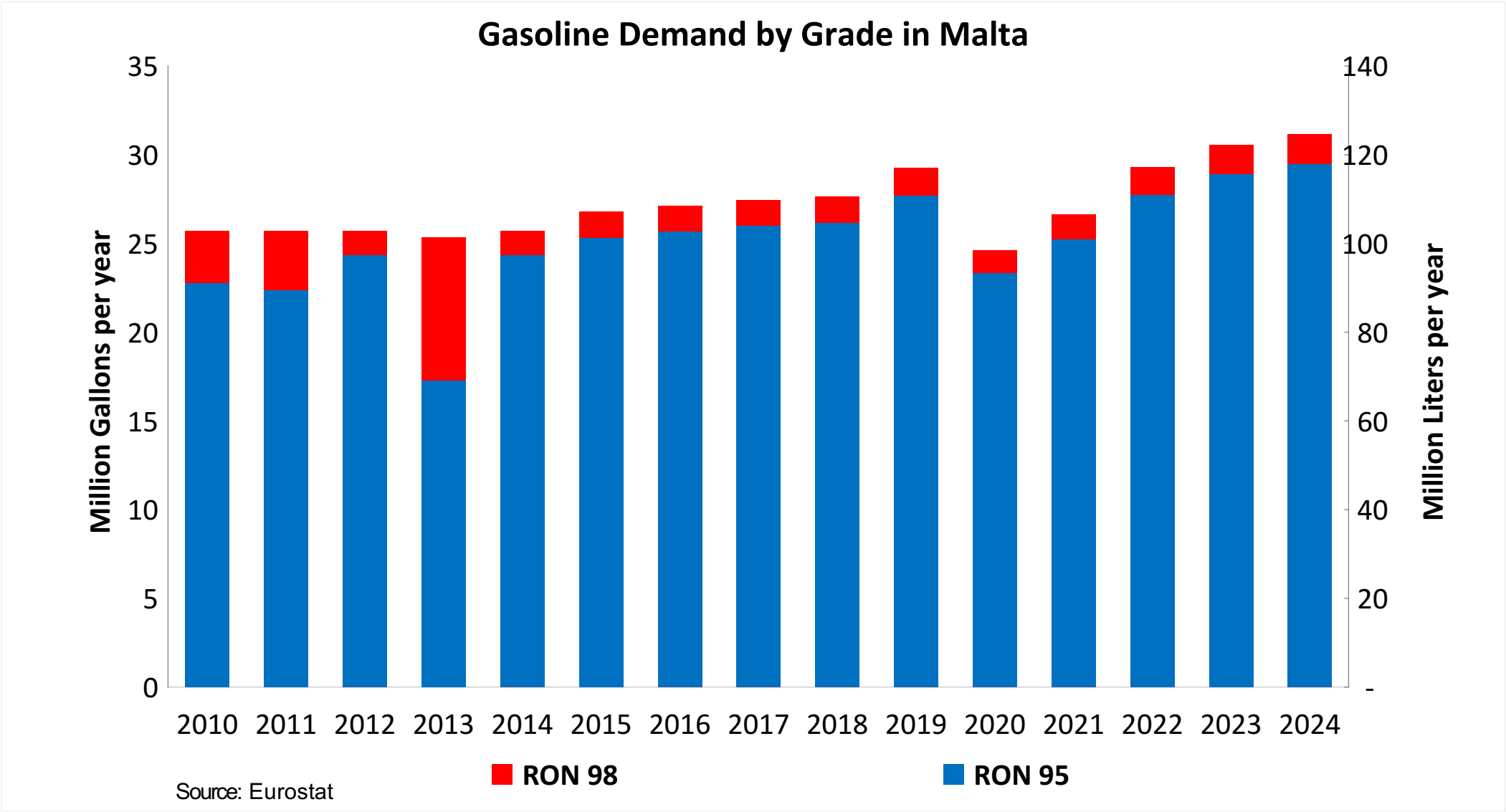


In 2024, gasoline consumption in Malta was 118 million liters (31 million gallons) and growth rate was 1.2%. There is no gasoline production. Gasoline imports were 163 million liters (43 million gallons). Malta complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.0 RON.

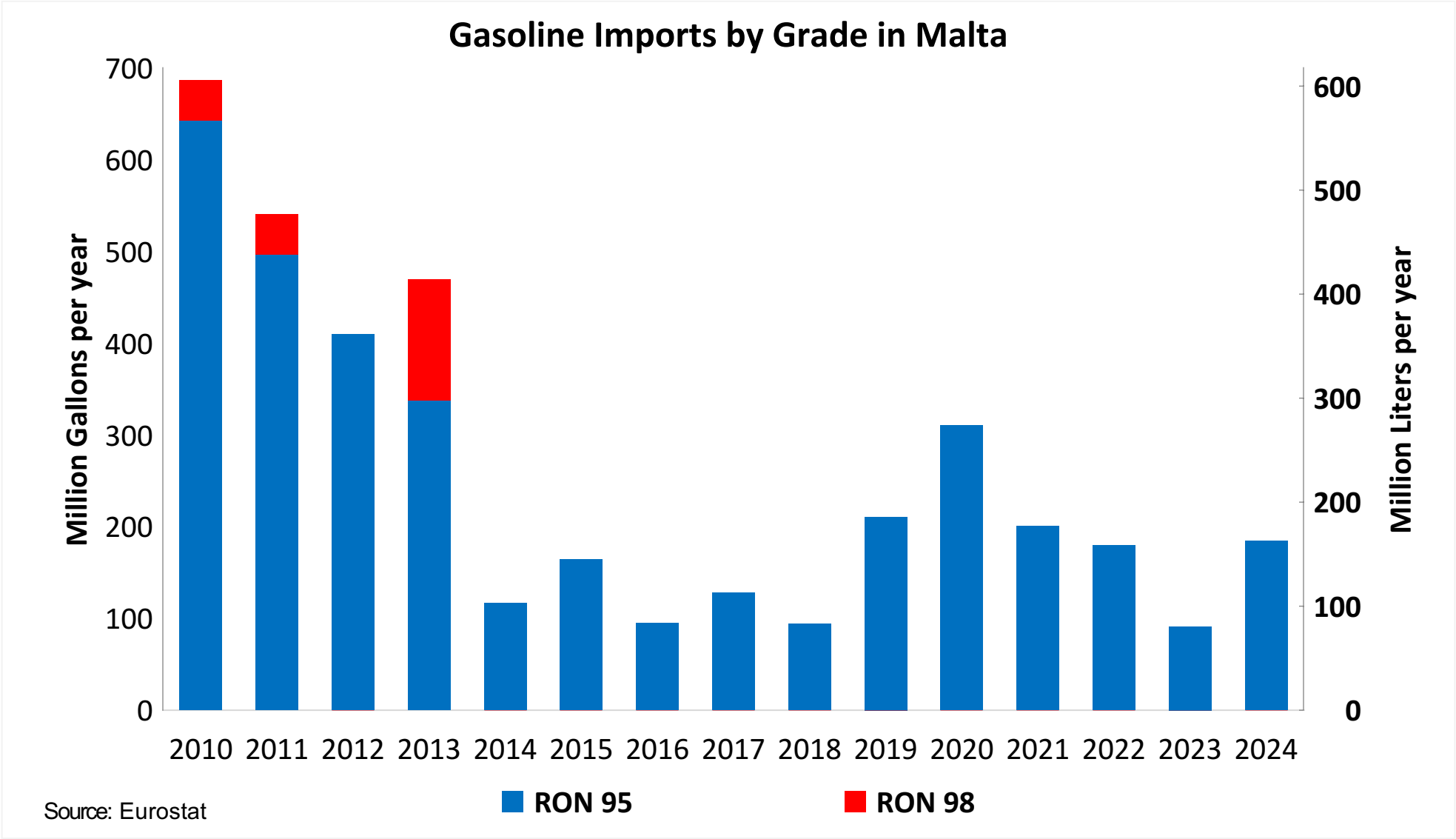
Malta does not produce or consume bioethanol at a commercial level

Source: European Commission, Eurostat

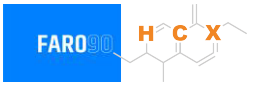
Gasoline Demand in Malta



Gasoline Imports in Malta



Ethanol Balance in Malta



Malta does not produce or consume bioethanol at a commercial level

Source: Eurostat

Gasoline Quality in Malta

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	0.2							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

The FARO90 logo is shown in a blue box, with the text "FARO90" in white. To the right of the logo is a chemical structure diagram of a branched alkene, with the letters "H", "C", and "X" in orange highlighting specific atoms in the structure.

Catalytic Gasoline
Light Primary Naphta
Heavy Primary Naphta
Reformate
Isopentane
Alkylate
Isomerate
Hydrocracked Gasoline
Coker Naphta
n-Butane
MTBE
ETBE
TAME
Aromatics

A horizontal stacked bar chart titled 'EU is a good idea' showing the percentage of respondents in various European countries who believe the EU is a good idea. The chart is divided into 12 color-coded segments representing different age groups. The countries are listed on the y-axis, and the x-axis shows the percentage from 0% to 100%.

Country	Overall % (Approx.)
Austria	95%
Belgium	96%
Bulgaria	98%
Croatia	99%
Cyprus	97%
Czechia	96%
Denmark	95%
Estonia	96%
Finland	96%
France	98%
Germany	97%
Greece	97%
Hungary	98%
Ireland	99%
Italy	99%
Latvia	97%
Lithuania	97%
Luxembourg	95%
Malta	99%
Netherlands	96%
Poland	98%
Portugal	97%
Romania	97%
Slovakia	96%
Slovenia	95%
Spain	97%
Sweden	95%
UK	99%

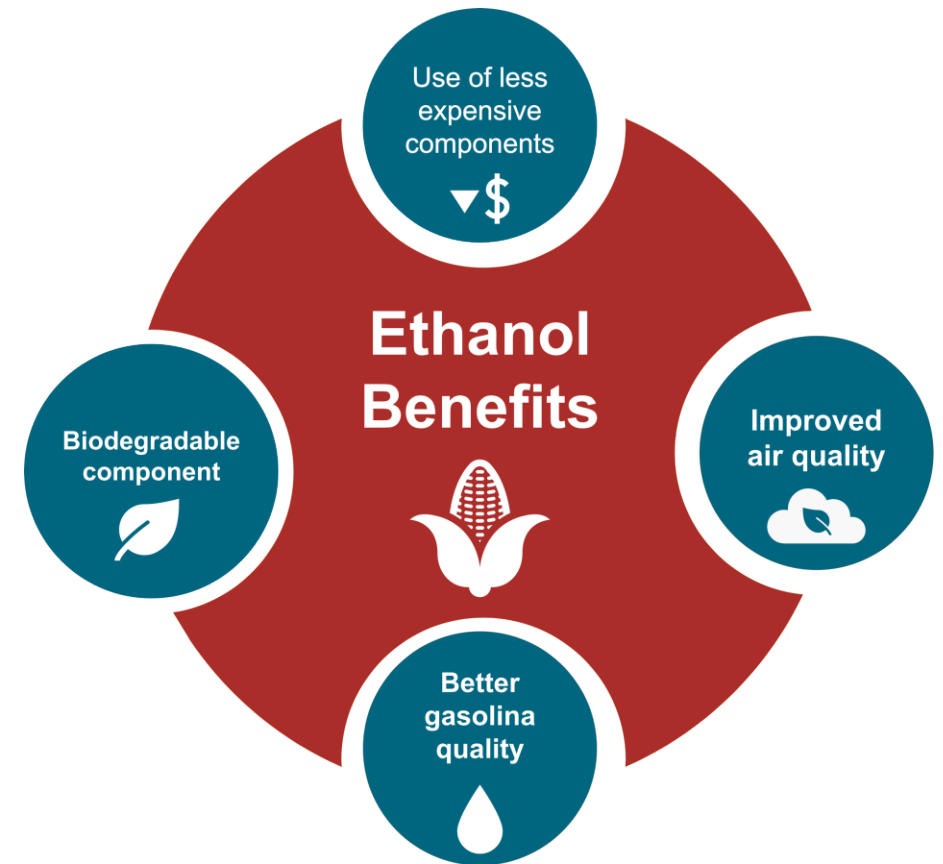
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

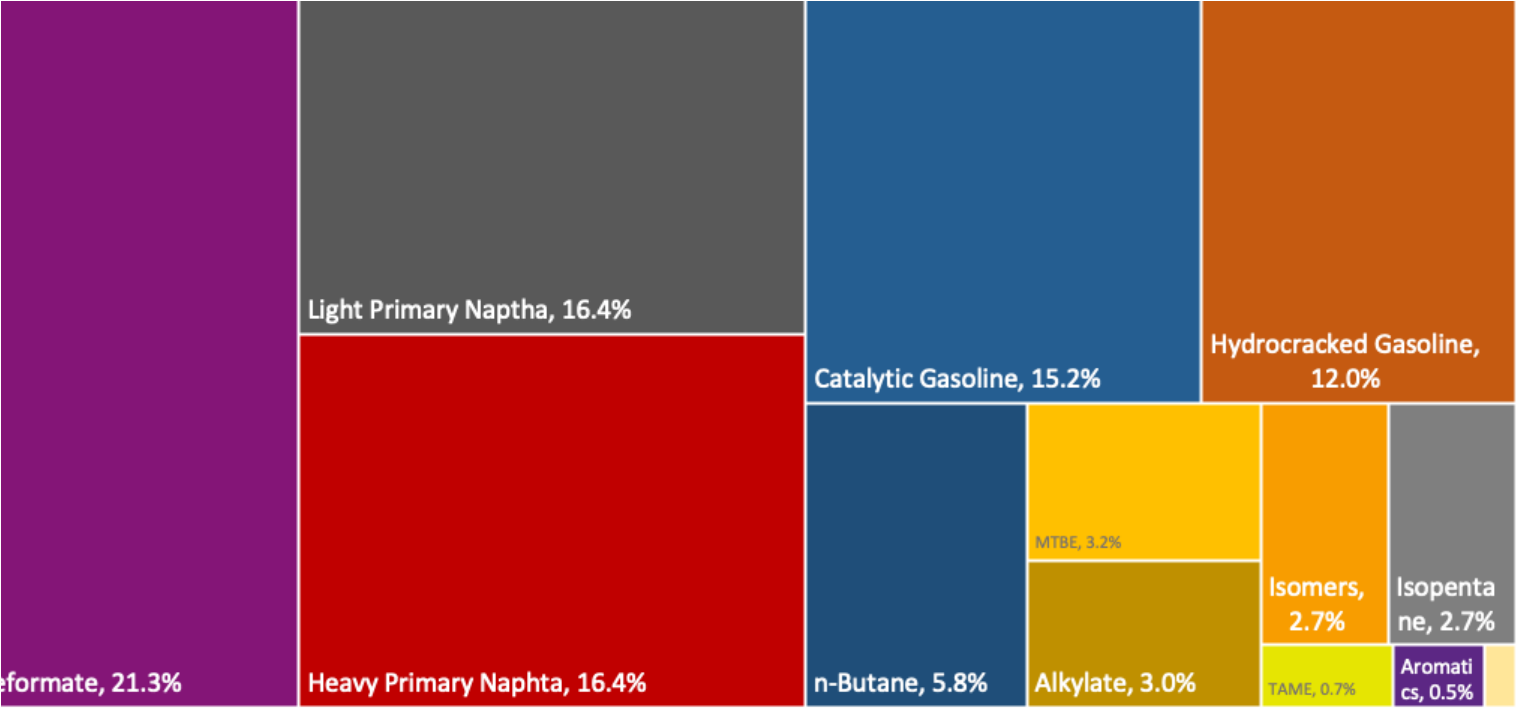
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

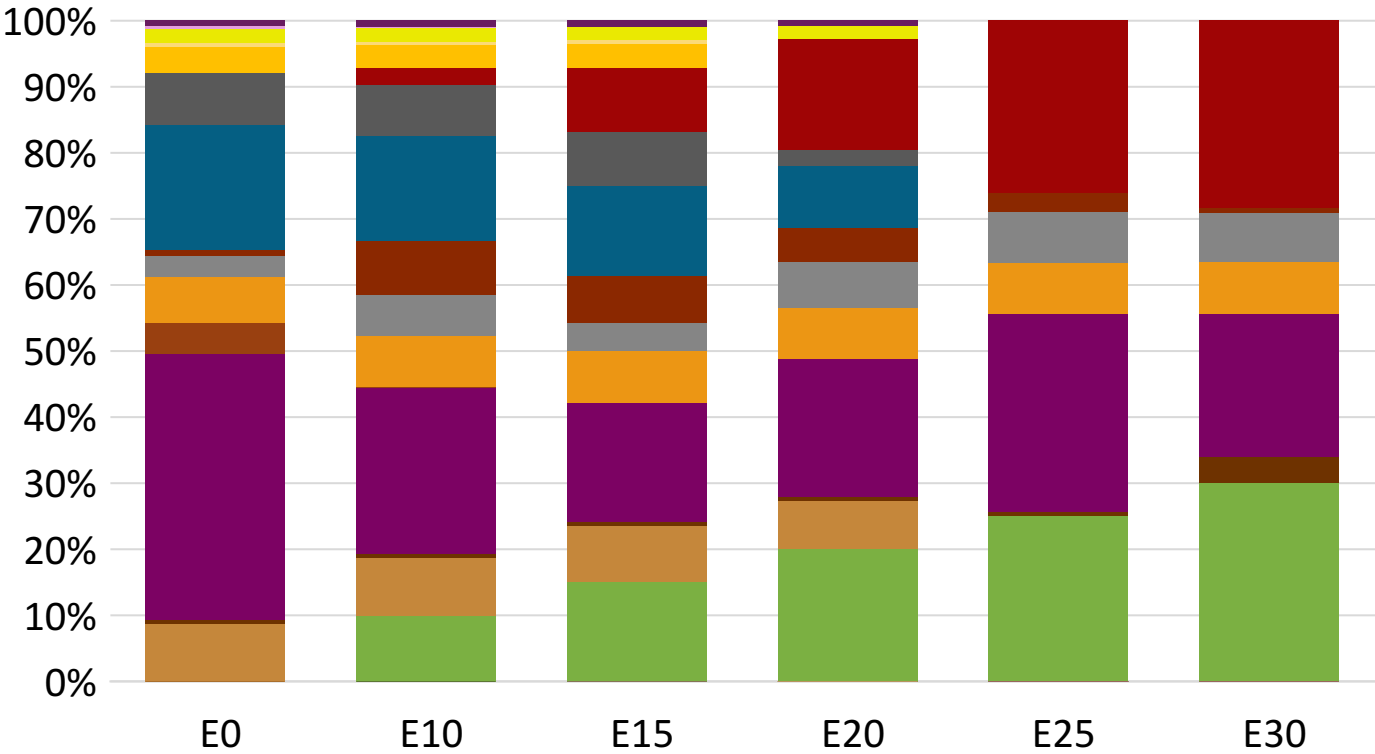
The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Malta Available Blending Components



Malta – Gasoline 95 – Constant Octane

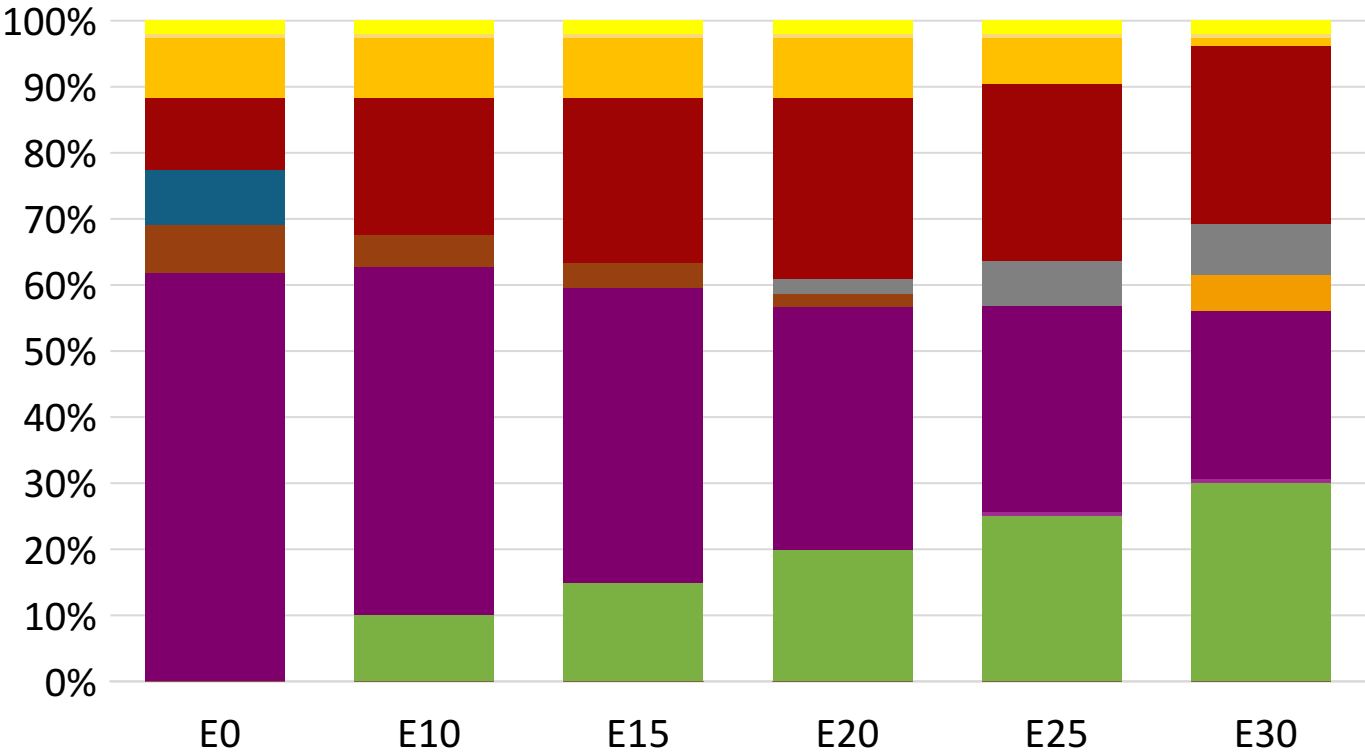


Average CIF 2024 Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.4	95.8
Price (US\$/gal)	\$ 2.258	\$ 2.145	\$ 2.062	\$ 1.986	\$ 1.891	\$ 1.817

Source: Faro90

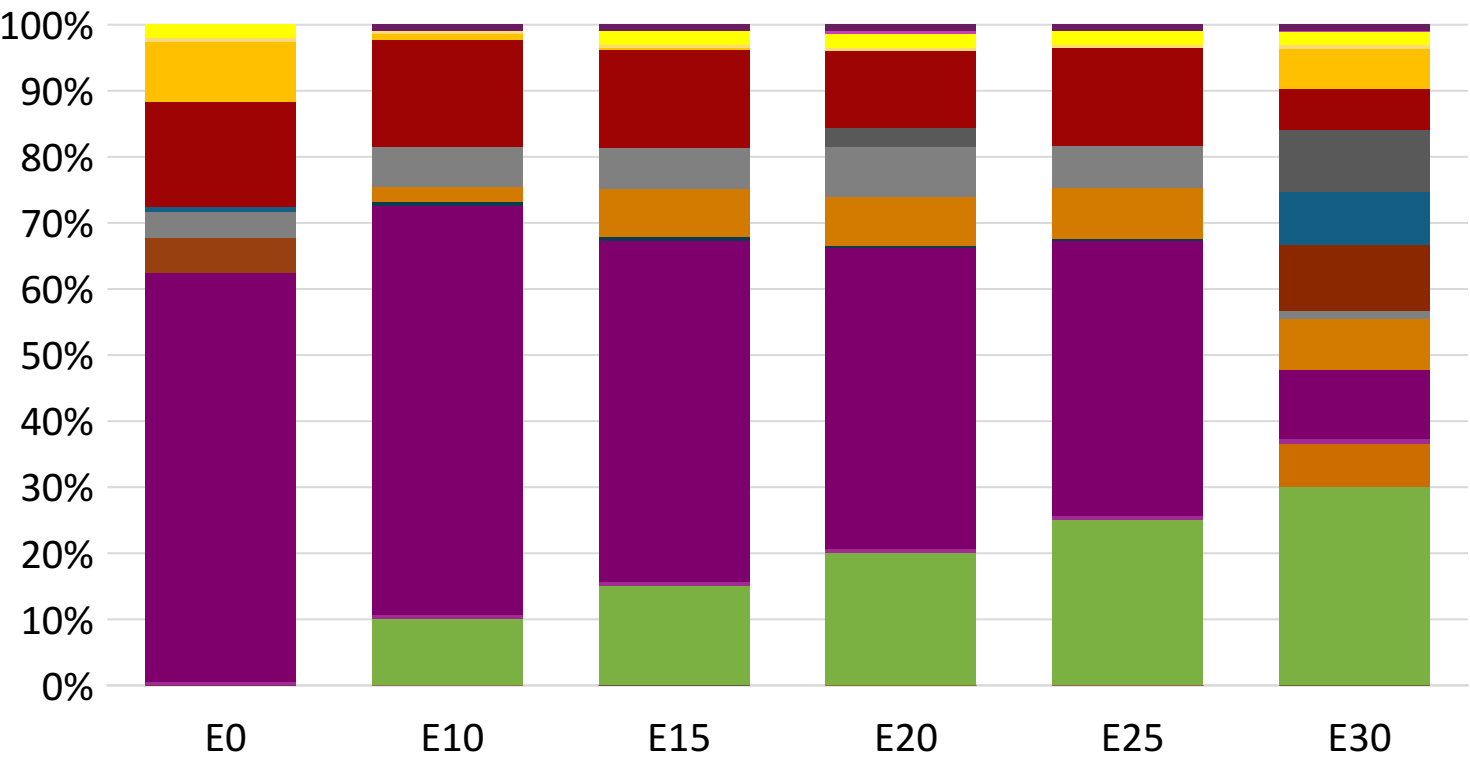
Malta - Gasoline 98 - Constant Octane



Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.307	\$ 2.188	\$ 2.119	\$ 2.049	\$ 1.979	\$ 1.926

Malta – Gasoline 95 – Increased Octane

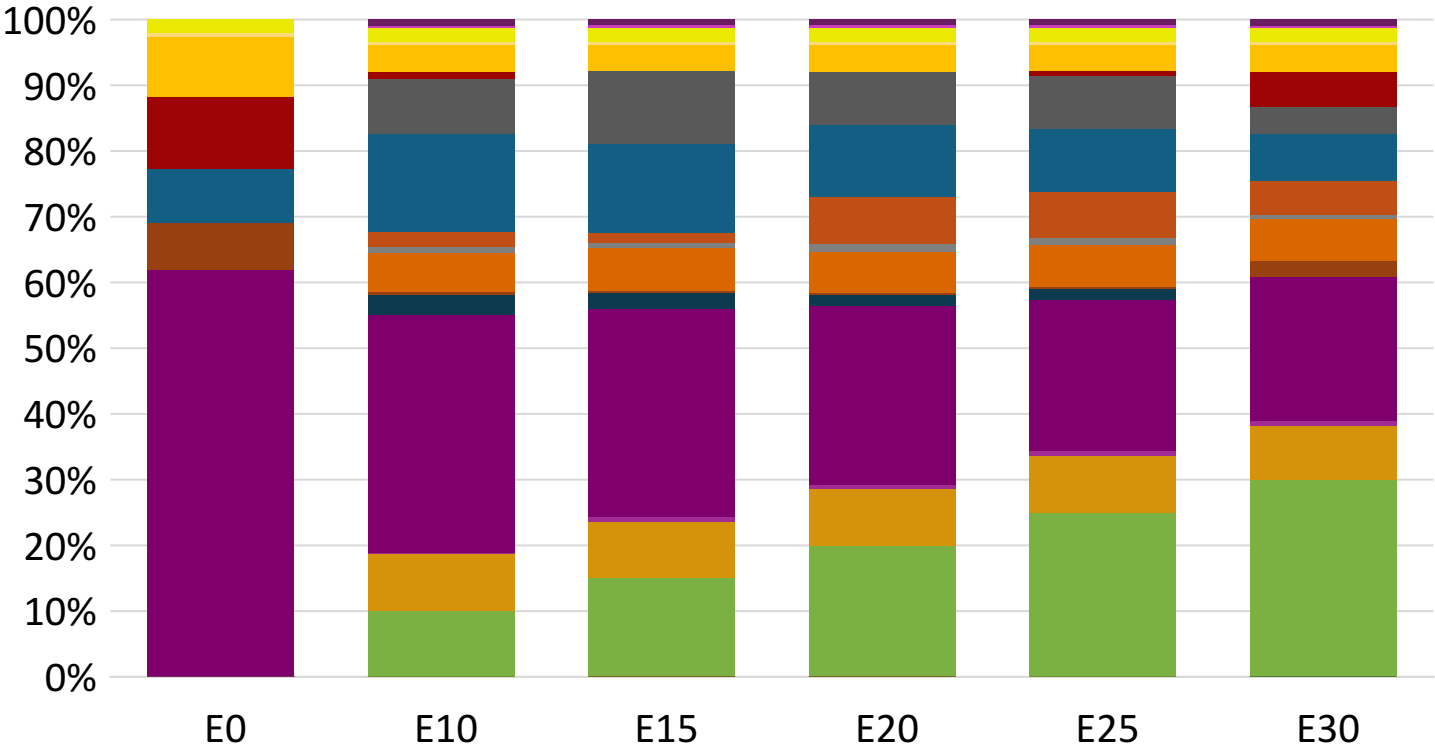


Average CIF 2024 Prices

Octane (RON)	95.0	95.9	97.5	99.0	100.5	101.8
Price (US\$/gal)	\$ 2.177	\$ 2.177	\$ 2.177	\$ 2.177	\$ 2.177	\$ 2.177

Source: Faro90

Malta – Gasoline 98 – Increased Octane



Average CIF 2024 Prices

Octane (RON)	98.0	98.8	100.1	101.3	102.6	104.1
Price (US\$/gal)	\$ 2.307	\$ 2.307	\$ 2.307	\$ 2.307	\$ 2.307	\$ 2.307

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

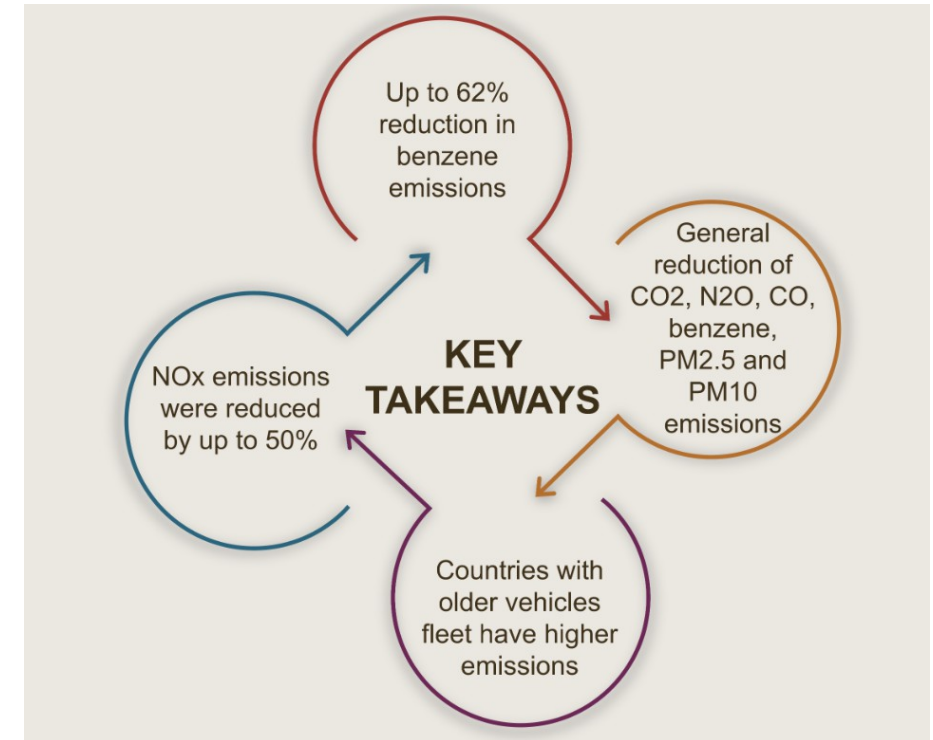
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

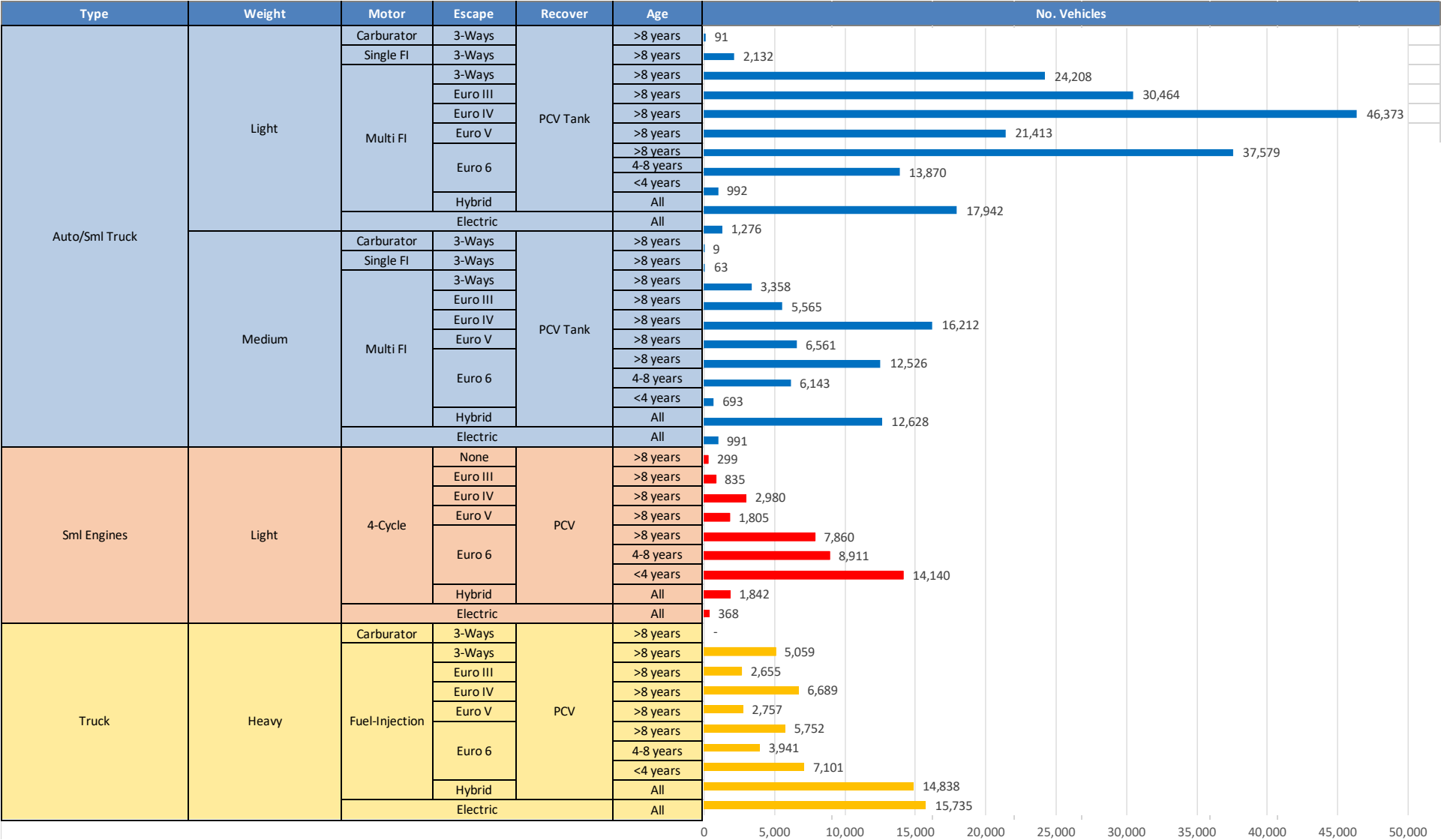
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



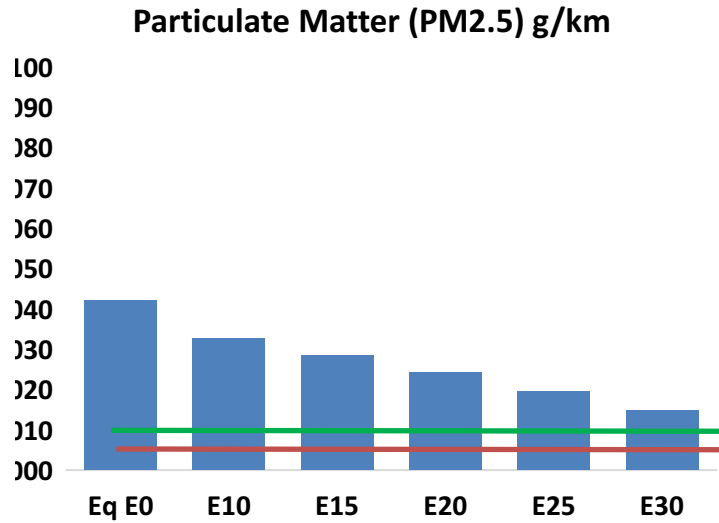
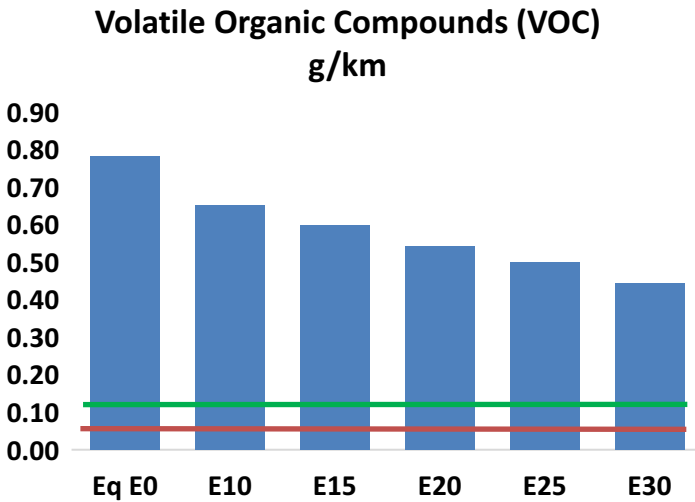
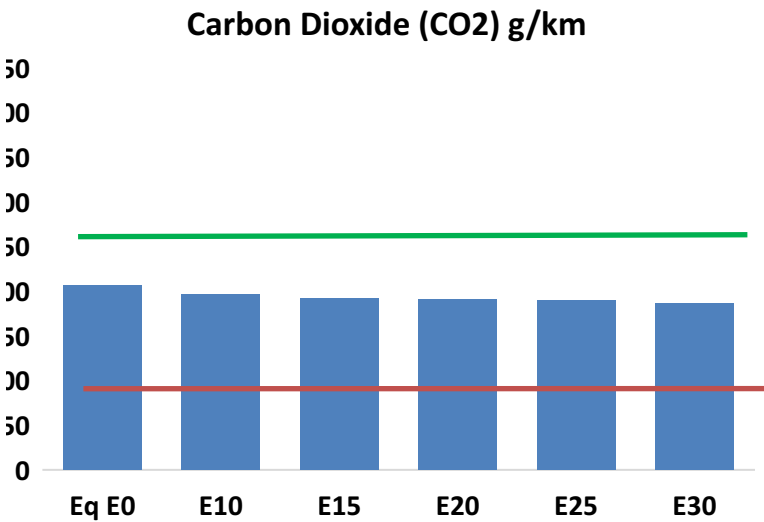
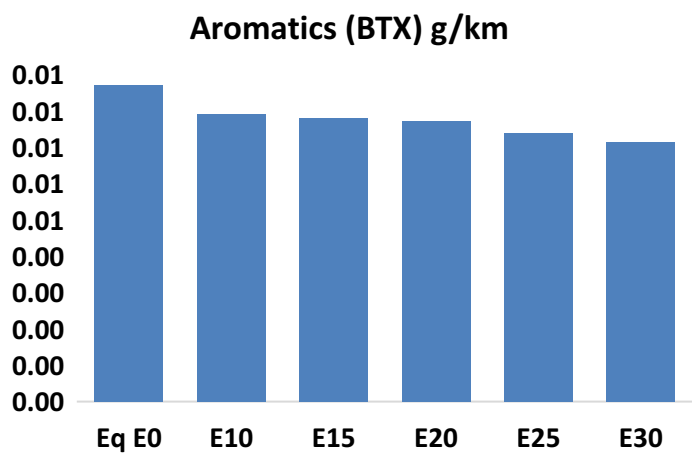
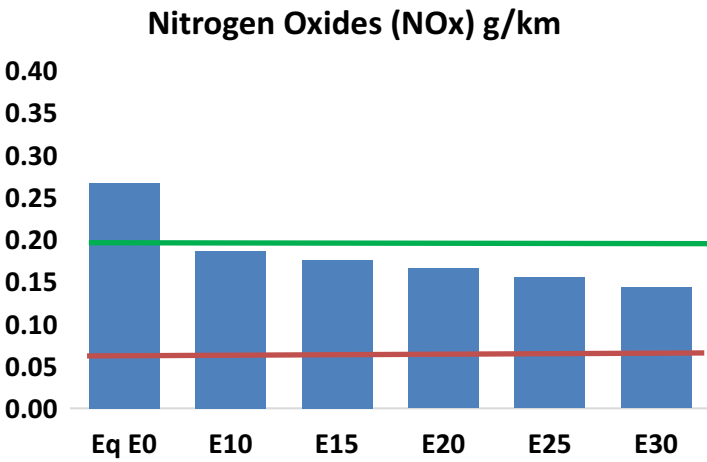
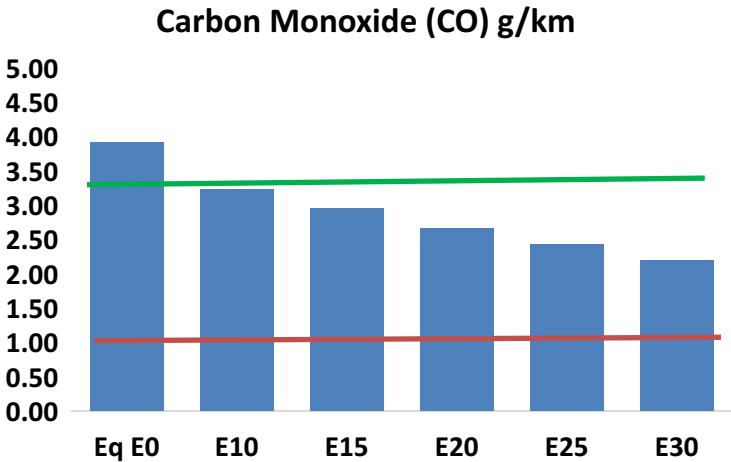
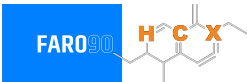
*Source: Bureau of transportation statistics.

Gasoline Vehicle Fleet - Malta



Malta – Vehicular Emissions

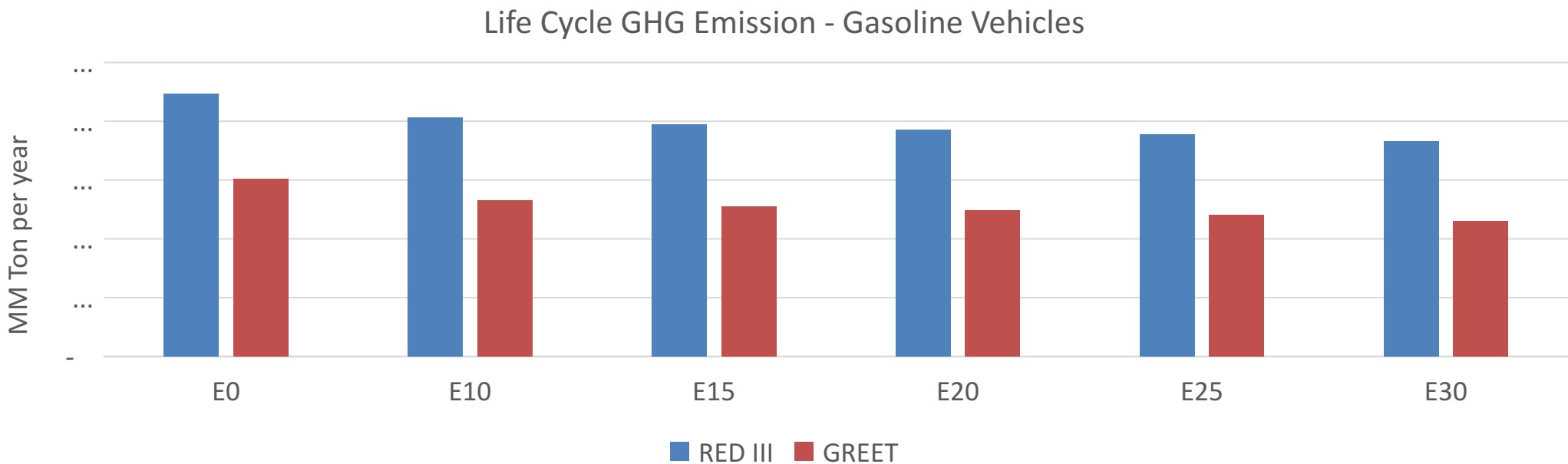
TIER III BIN 125 United States
Euro 6 Standard



Malta – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	19,908	18,177	16,446	14,988	13,502	12,355	11,130	-17%	-32%
CO2	1,052,550	1,025,484	999,922	979,818	969,870	964,236	946,463	-5%	-8%
VOC	3,980	3,647	3,315	3,038	2,756	2,539	2,252	-17%	-31%
NOx	1,355	1,016	948	894	843	787	728	-30%	-38%
SOx	13	12	11	10	9	8	7	-15%	-28%
NH3	404	400	396	398	402	405	408	-2%	0%
N2O	15	15	15	15	16	16	16	-1%	2%
CH4	807	807	807	819	837	852	867	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	20	19	18	18	17	17	17	-8%	-13%
Acetaldehyde	66	76	111	169	230	263	310	68%	249%
Formaldehyde	245	238	277	326	341	378	411	13%	39%
Benzene	44	43	40	40	39	38	36	-9%	-11%
PM2.5	66	59	51	45	38	31	23	-22%	-43%
PM10	148	131	115	99	84	69	52	-22%	-43%

Malta – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	2.2
Target @ 2035 (MMT/year)	1.2
Delta	1.0

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.4%	15.6%	18.1%	20.5%	23.7%
RED II/III	0.0%	9.2%	11.7%	13.7%	15.6%	18.1%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	7.8%	9.8%	11.4%	12.9%	15.0%
RED II/III	0.0%	8.5%	10.9%	12.8%	14.6%	16.9%